Lab-7

Task-1

1 2 3 4 5

Q=1 {1, 2} size = 2,

Q=2 {3, 4} size = 2,

Q=3 = {1, 2, 3, 4} size = 4,

We used a function union to combine the circles. ~~If~~ The function will call representative which ~~does~~ gives us the parents which means with whom the person in friends with. If u and v is not the same person then it adds to the circle and updates it's parent.

Task-2

This code is like a weighted graph so we create a adjacency list and a visited list to traverse the nodes. To count the minimum cost we took a priority queue to store the explored in the order of their maintenance costs. At first pop the city and check if it's visited. then it adds the weight to the cost. and check it's adjacent. It's almost like a BFS function.

Task-3

The array here will each element represents the number of distinct ways Freddy can climb to the desired step. The jump count counts the number of jumps. The base case is zero. If n is 1 the it can jump 1. If it's ~~to~~ 2 then two and so on.

Task-4

Here type represents different the number of ~~coin~~ coin type ~~direction~~. and our target is the amount of money. There can be three base case. If it's 0 then the coin is 0. If it's below zero that is impossible and If it's between 0 and 1. it's also impossible. The recursive call calculates the minimum number of coins where it recursively calls the previous type. At last if the result is infinity then prints (-1) and if not the prints the minimum coin value.